

Martingales Convergence Theorems

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Abstract

This paper explores the utility of Martingale Convergence Theorems in probability theory, including its proof. The Forward Convergence Theorem is first highlighted, followed by a discussion on the certainty of gambler's ruin in unfavorable games using the theorem. Subsequently, the paper discusses the Backward Convergence Theorem, which is shown to be useful for establishing the all-or-nothing outcome of exchangeable sequence of events as described by the Hewitt-Savage 0–1 Law. Finally, the paper shows the usage of the Backward Martingale convergence theorem for proving the Strong Law of Large Numbers (SLLN).

1 Introduction

The term martingale has an important history before entering mathematical vocabulary. Originally borrowed from gambling terminology, where it described a betting strategy of doubling one's stake after each loss, the word was formally introduced into probability theory by Jean Ville in his 1939 thesis. The latter explicitly stated that he borrowed the name directly from gamblers' vocabulary, as probabilists of that era frequently encountered individuals claiming to possess winning strategies like "martingales" for games like roulette. The mathematical content that this concept unleash predates its formal naming. The mathematical property that defines it was first implicitly used in Pascal's solution to the famous Problem of Points in 1654. The English adoption of the term came through Doob, who reviewed Ville's thesis and subsequently used martingale in his classic book *Stochastic Processes*.

In the mathematical vocabulary, Martingales are integrable stochastic processes satisfying a particular conditional expectation property: the expected future value, given all past information, equals the present value. The importance of martingales extends far beyond theoretical interest, in financial mathematics, the fundamental theorem of asset pricing establishes that a financial market is viable (free of arbitrage opportunities) if and only if there exists a probability measure under which asset risk prices form martingales. Beyond finance, martingale theory finds other applications in diverse areas including the analysis of algorithms, branching processes in population genetics, optimal stopping problems, statistical sequential ones ¹, and the study of harmonic functions through probabilistic methods ².

¹Due to the ability of martingales to account for random variables that are dependent.

²The martingale provides the equivalent of the mean value property.

Limiting conditions on this stochastic process can allow for the future outcome to be approaching a fixed value. This roughly what the martingale convergence theorems aims to establish, which are among the most powerful and widely applicable results in probability theory. The forward martingale convergence theorem establishes conditions under which a martingale converges almost surely as time progresses to infinity. This result is widely used to prove almost sure convergence in diverse fields from mathematical finance to statistical physics ³, and it establishes the certainty of gambler's ruin in unfavorable games. The backward martingale convergence theorem addresses martingales indexed in reverse time order. This theorem establishes a very quick proof of the strong law of large numbers and the Hewitt-Savage 0 – 1 Law.

2 Preliminaries

Definition 2.1. A real valued process $(W_n)_{n \geq 0}$ is a **martingale** relative to a process $(X_n)_{n \geq 0}$ if

- (i) W_n is uniquely determined by $X_0, X_1, \dots, X_n$
- (ii) $\mathbb{E}|W_n| < \infty$ (existence of expectation)
- (iii) $\mathbb{E}[W_{n+1}|X_0, \dots, X_n] = W_n$ (fairness assumption)

For a **super-martingale**, we replace the third property with

- (iii)* $\mathbb{E}[W_{n+1}|X_0, \dots, X_n] \leq W_n$

While for **sub-martingale**, the third property becomes

- (iii)** $\mathbb{E}[W_{n+1}|X_0, \dots, X_n] \geq W_n$

Definition 2.2. A real-valued process $(W_n)_{n \leq 0}$ is a **backward martingale** relative to a process $(X_n)_{n \leq 0}$ if:

1. W_n is uniquely determined by $X_n, X_{n-1}, \dots$
2. $\mathbb{E}[|W_n|] < \infty$ for all $n \leq 0$.
3. $\mathbb{E}[W_{n+1} | X_n, X_{n-1}, \dots] = W_n$.

Definition 2.3. Let $(\Omega, \mathcal{F}, P)$ be a probability space. A sequence of σ -algebras $(\mathcal{F}_n)_{n \in T}$ is called a **filtration** if it satisfies:

1. Each $\mathcal{F}_n$ is a sub- σ -algebra of $\mathcal{F}$ ($\mathcal{F}_n \subseteq \mathcal{F}$).
2. The sequence is nested (increasing): for all $n < m$ in T , we have $\mathcal{F}_n \subseteq \mathcal{F}_m$.

tentative explanation: An intuitive way of thinking of filtrations is viewing them as a process that also account for accumulation.

Definition 2.4 (Definition of a Backward Martingale using filtrations). A sequence of random variables $(W_n)_{n \leq 0}$ is called a **Backward Martingale** (Backward MG) with respect to the filtration $(\mathcal{F}_n)_{n \leq 0}$ if the following conditions hold:

1. **Filtration Structure:** The sequence of σ -algebras $(\mathcal{F}_n)_{n \leq 0}$ is increasing in the index n :

$$\dots \subset \mathcal{F}_{-3} \subset \mathcal{F}_{-2} \subset \mathcal{F}_{-1} \subset \mathcal{F}_0$$

³Boltzmann equation and Nanbu-type Monte Carlo schemes.

2. **Adaptivity:** X_n is $\mathcal{F}_n$ -measurable (adapted) for all $n \leq 0$.
3. **Integrability:** $\mathbb{E}[|X_n|] < \infty$ for all $n \leq 0$.
4. **Martingale Property:**

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n, \quad \text{for all } n < 0$$

3 Forward Convergence Theorem

Theorem 3.1 (Doob's Martingale Convergence Theorem). *Suppose $(W_n)_{n \geq 0}$ is a super-martingale such that*

$$\sup_n (\mathbb{E}|W_n|) < \infty$$

Then, for $W = \limsup W_n$ ⁴, W_n converges to W almost surely where $\mathbb{E}|W| < \infty$.

tentative explanation: This statement is saying that almost all trajectories of a martingale converge to a certain value though it is not the same value for all trajectories. Next, we will prove two useful Lemma's for the proof of this theorem. We will add new definitions as needed.

Definition 3.1. *Consider a martingale $(W_n)_{n \geq 0}$. For $a < b$, let $U_N[a, b]$ be the number of upcrossings of $[a, b]$ until time N . An **upcrossing** occurs when, for a given trajectory $W_n(\omega)$, there exists $s < t$ such that $W_s(\omega) < a < b < W_t(\omega)$. Assuming that the zone $[a, b]$ inherits an order from the set of all possible outcomes, $U_N[a, b](\omega) = k$ is the largest value such that: $0 \leq s_1 < t_1 < s_2 < t_2 < \dots < s_k < t_k \leq N$*

$$W_{s_i}(\omega) < a < b < W_{t_i}(\omega), \quad i = 1, \dots, k.$$

Definition 3.2. *A sequence of random variables $\{C_n\}_{n \geq 1}$ is predictable with respect to its process if, for every time $n \geq 1$, the value of C_n is a function of the history of the X process up to time $n - 1$:*

$$C_n = g(C_1, \dots, C_{n-1})$$

where g is a non-random, measurable function. (Durrett, p.219)

Now, we will prove two useful Lemma's which will allow us to prove Theorem 3.1.

Lemma 3.1 (Doob's Upcrossing). *Let $(W_n)_{n \geq 0}$ a super-martingale. For $a < b$ and time N ,*

$$\mathbb{E}[U_N[a, b]] \leq \frac{\mathbb{E}[W_N^{(a-)}]}{(b - a)} \tag{1}$$

where

$$W_N^{(a-)} = \begin{cases} a - W_N & \text{if } W_N \leq a \\ 0 & \text{otherwise} \end{cases}$$

⁴Recall that the limsup of a sequence of real numbers always exists.

Proof. Consider the predictable sequence $(C_n)_{n \geq 1}$ such that

$$C_1(\omega) = \begin{cases} 1 & \text{if } W_0(\omega) < a \\ 0 & \text{otherwise} \end{cases}$$

Then, for $n \geq 2$

$$C_n(\omega) = \begin{cases} 1 & \text{if } C_{n-1}(\omega) = 1 \text{ and } W_{n-1}(\omega) \leq b \\ 1 & \text{if } C_{n-1}(\omega) = 0 \text{ and } W_{n-1}(\omega) < a \\ 0 & \text{otherwise} \end{cases}$$

Some observations on (C_n) :

- if $W_0 < a$, then $C_1 = 1$ and C_n remains one until $W_n > b$.
- C_n remain zero until $W_n < a$
- if $W_0 > a$, then $C_1 = 0$ and it remains zero until $W_n < a$.
- C_n remains one until $W_n > b$

Now, we will define a new process $(Y_n)_{n \geq 1}$ such that

$$Y_n = (C \cdot W)_n^5 = \sum_{1 \leq j \leq n} C_j(W_j - W_{j-1})$$

Claim 1: $Y_N(\omega) \geq (b - a)U_N[a, b] - W_N^{(a^-)}(\omega)$,

where $U_N[a, b] = k$ and s_i, t_i , for $i = 1, \dots, k$, are defined as in definition 3.1.

Next, we can make some observations on the values of C_n between the times s_i and t_i that correspond to an upcrossing of $[a, b]$. Recall that $W_{s_i}(\omega) < a < b < W_{t_i}(\omega)$.

First, independent of C_{s_i} , $C_{s_i+1} = 1$ since $W_{s_i}(\omega) < a < b$.

More generally, we notice that $C_t(w) = 1$ for $s_i + 1 \leq t \leq l_i$, where $l_i \leq t_i$ is the smallest time (greater than s_i) such that $W_t(\omega) > b$. Wlog, let $s_1 = \min \{n : W_n(\omega) < a\}$ and $s_{j+1} = \min \{n > t_j : W_n(\omega) < a\}$ for $j = 1, \dots, k$. Next, we will rewrite $Y_N(\omega)$ ⁶ in terms of $C_t(\omega)$

$$\begin{aligned} Y_N &= \sum_{1 \leq j \leq N} C_j(W_j - W_{j-1}) \\ &= \sum_{t < s_1} C_{t+1}(W_{t+1} - W_t) + \sum_{1 \leq i \leq k} \left(\sum_{s_i \leq t < l_i} C_{t+1}(W_{t+1} - W_t) + \sum_{l_i \leq t < s_{i+1}} C_{t+1}(W_{t+1} - W_t) \right) \\ &\quad + \sum_{t \geq s_{k+1}} C_{t+1}(W_{t+1} - W_t) \\ &= \sum_{1 \leq i \leq k} (W_{l_i} - W_{s_i}) + W_N - W_{s_{k+1}} \\ &= \sum_{1 \leq i \leq k} (W_{l_i} - W_{s_i}) - (W_{s_{k+1}} - W_N) \end{aligned}$$

⁵This product is called a stochastic integral. In our case, it is discrete.

⁶Although the calculation is done for a specific trajectory, I am dropping the ω 's.

where the first summation and the second term in the second summation always die out. Moreover, the last summation would also be zero if $s_{k+1} > N$. Notice that

$$\sum_{1 \leq i \leq k} (W_{l_i} - W_{s_i}) \geq k(b - a) = U_N[a, b](b - a)$$

Additionally, if $a < W_{s_{k+1}} \leq W_N$, then

$$W_{s_{k+1}} - W_N \leq 0$$

If $W_N \leq a < W_{s_{k+1}}$, then we just take

$$W_{s_{k+1}} - W_N \leq a - W_N$$

Then, we have that $(W_{s_{k+1}} - W_N) \leq W_N^{(a^-)}$.

Hence,

$$Y_N \geq U_N[a, b](b - a) - W_N^{(a^-)} \quad (2)$$

Claim 1 is verified.

Claim 2: $(Y_n)_{n \geq 1}$ is a super martingale with $\mathbb{E}[Y_N] \leq \mathbb{E}[Y_0] = 0$. We will take this claim for granted.

Finally, take the expectation on both sides of inequality (2), we get our Lemma. $\square$

Lemma 3.2. *Consider the conditions of Theorem 3.1. For $a < b$, let*

$$\Lambda_{a,b} = \{\omega : \lim_{N \rightarrow \infty} U_N[a, b] = \infty\}.$$

The probability of trajectories having infinitely many upcrossings of $[a, b]$ is zero. In other words,

$$\mathbb{P}(\Lambda_{a,b}) = 0.$$

Proof. By Lemma 3.1,

$$\mathbb{E}[U_N[a, b]] \leq \frac{1}{b - a} \mathbb{E}[W_N^{a^-}] \leq \frac{1}{b - a} (\sup_N \mathbb{E}|W_N| + a) < \infty,$$

where the second inequality is due to triangular inequality and the finiteness on RHS due to the assumptions of our theorem. Now, note that $U_N[a, b]$ is a non-decreasing R.V. (for each trajectory, it is non-decreasing), so it converges to $\lim_{N \rightarrow \infty} U_N[a, b]$. Hence, by Monotone convergence theorem⁷, the limit of the expectation is the expectation of the limit. We know that the former converges and is finite, so $\mathbb{E}[\lim_{N \rightarrow \infty} U_N[a, b]]$ is also finite. Finally, this implies $\mathbb{P}(\lim_{N \rightarrow \infty} U_N[a, b] = \infty) = 0$. $\square$

Now, we are ready to prove Doob's convergence theorem.

Theorem 3.1 proof. Consider the set of trajectories that diverge

$$\begin{aligned} \Lambda &:= \{\omega : W_n(\omega) \text{ such that the limit does not exist}\} \\ &= \{\omega : \liminf_n W_n(\omega) < \limsup_n W_n(\omega)\}^8 \\ &= \bigcup_{a < b, a, b \in \mathbb{Q}} \{\omega : \liminf_n W_n(\omega) < a < b < \limsup_n W_n(\omega)\}, \end{aligned}$$

⁷Measure theoretic MCT

the last equality of sets is essentially taking the union of all trajectories where the distance between the limsup and liminf is atleast $\epsilon > 0$ (rational values, which are dense). The set can also be reinterpreted in terms of upcrossings.

Notice that $\liminf_n W_n(\omega) < a < b < \limsup_n W_n(\omega)$ implies that for this given trajectory the values of $W_n(w)$ will keep alternating between $W_n(w) > b$ and $W_n(w) < a$, i.e. we have infinitely many crossings over $[a, b]$. Hence,

$$\begin{aligned}\Lambda &= \bigcup_{a < b, a, b \in \mathbb{Q}} \{\omega : \liminf_n W_n(\omega) < a < b < \limsup_n W_n(\omega)\} \\ &= \bigcup_{a < b, a, b \in \mathbb{Q}} \Lambda_{a,b}\end{aligned}$$

By Lemma 3.2, and the fact that union of measure zero sets has zero measure, $\mathbb{P}(\Lambda) = 0$. Therefore,

$$W_n \rightarrow W \quad (\text{almost surely}),$$

for a random variable W .

Next, we look at the existence of expectation of W

$$\begin{aligned}\mathbb{E}|W| &= \mathbb{E}[\liminf_n |W_n|] \quad (W = \liminf_n |W_n| \text{ a.s.}) \\ &\leq \liminf_n \mathbb{E}|W_n| \quad (\text{Fatou's Lemma}) \\ &\leq \sup_n \mathbb{E}|W_n| < \infty \quad (\sup_n \inf_{k \geq n} \{\} < \sup_n \sup_{k \geq n} \{\} = \sup_n \{\})\end{aligned}$$

The proof of our theorem is complete. $\square$

Example 3.1 (Application: Gambler's ruin). *Consider the unfavorable/fair case $p \leq 1/2$ of winning \$1, where the gambler starts with initial capital $X_0 = i > 0$ against infinite capital.*

1. **Supermartingale Property:** *The gambler's fortune X_n is a non-negative process that satisfies the supermartingale condition.*

$$\begin{aligned}E[X_{n+1}|X_0, \dots, X_n] &= p(X_n + 1) + (1 - p)(X_n - 1) \\ &= X_n + (2p - 1)\end{aligned}$$

$$(2p - 1) \leq 0. \text{ Thus, } E[X_{n+1}|X_0, \dots, X_n] \leq X_n.$$

2. **Bounded in L^1 :** *Using the law of total expectation: $E[X_n] \leq E[X_0] = i$. Therefore, $\sup_n (E|X_n|) \leq i < \infty$.*
3. **Almost Sure Convergence:** *By Doob's Martingale Convergence Theorem, the supermartingale X_n converges almost surely to a finite random variable X : $X_n \rightarrow X$.*
4. **Almost Sure Ruin:** *X_n is viewed as a random walk with zero or negative drift, an absorbing state 0 ⁹, and it cannot escape to infinity. For $X > 0$ to hold with positive probability, the random walk would have to eventually stop moving against its negative/zero drift, which is impossible. Therefore, the only possible finite limit is $X = 0$.*

Thus, the probability of ultimate ruin is $P(\lim_{n \rightarrow \infty} X_n = 0) = 1$ when $p \leq 1/2$.

⁹no upper absorbing barrier

4 Backward Convergence Theorem

Theorem 4.1 (Backward Martingale Convergence Theorem). *Let $(W_n)_{n \leq 0}$ be a Backward Martingale as defined in Definition 2.3 . Then*

$$\lim_{n \rightarrow -\infty} W_n = W \quad \text{exists almost surely and in } L^1.^{10}$$

Note: The proof of this version of the Backward convergence theorem follows exactly the same steps of proving Theorem 3.1.

Note: In this case, it's guaranteed to have

$$\sup_n (\mathbb{E}[|W_n|]) < \infty.$$

This is because due to Jenson's inequality and the law of total expectation, we have

$$\sup_{n \leq 0} \mathbb{E}[|W_n|] = \mathbb{E}[|W_0|].$$

Furthermore, the sequence $(W_n)_{n \geq 0}$ is uniformly integrable ¹¹, which means means the sequence of random variables is well-behaved in its tails, ensuring that the expected contribution from extreme values uniformly vanishes. The backward martingale satisfies this because every term W_n is a conditional expectation of the single integrable random variable W_0 , a structure which guarantees the uniform vanishing of the tails due to the integrability of W_0 . This automatic UI condition is what directly guarantees the stronger L^1 convergence for the backward martingale.

Example 4.1 (Hewitt-Savage 0-1 Law). *Let $(\xi_i)_{i \geq 1}$ be i.i.d. random variables and let $\mathcal{E}$ be the exchangeable σ -algebra. Then for every $A \in \mathcal{E}$ ¹²,*

$$\mathbb{P}(A) \in \{0, 1\}.$$

Only a Sketch of the proof to show the role of the backward martingale convergence theorem: Let A be an event in the exchangeable σ -algebra. Consider the decreasing family of σ -algebras

$$\mathcal{F}_{-n} := \sigma(\xi_{n+1}, \xi_{n+2}, \dots),$$

Define

$$W_{-n} := \mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_{-n}].$$

The sequence $(W_{-n})_{n \geq 1}$ is a backward martingale. By the backward martingale convergence theorem, W_{-n} converges almost surely to

$$\mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_{-\infty}] = \mathbb{I}_A, \quad \mathcal{F}_{-\infty} := \bigcap_n \mathcal{F}_{-n},$$

Now, it can be shown that

$$A \in \mathcal{F}_{-\infty}$$

, then using Kolmogorov's 0-1 Law, we get $P(A) \in \{0, 1\}$.

¹⁰

$$W_n \xrightarrow{L^1} W \quad \text{means that} \quad \lim_{n \rightarrow -\infty} \mathbb{E}[|W_n - W|] = 0.$$

¹¹ $\lim_{a \rightarrow \infty} (\sup_n \mathbb{E}[|W_n| \cdot \mathbf{1}_{\{|W_n| > a\}}]) = 0$

¹² An event A is in $\mathcal{E}$ if and only if the event A remains the same even if you exchange a finite number of the random variables X_n .

Example 4.2 (SLLN Using Backward Martingale). *Let $(\xi_i)_{i \geq 1}$ be i.i.d. random variables with $\mathbb{E}|\xi_1| < \infty$. Then*

$$\frac{1}{n} \sum_{i=1}^n \xi_i \xrightarrow{a.s.} \mathbb{E}[\xi_1].$$

Proof. Let $X_{-n} = \frac{S_n}{n}$ where $S_n = \xi_1 + \xi_2 + \dots + \xi_n$ and $F_n = \sigma(\mathcal{S}_n, \xi_{n+1}, \xi_{n+2}, \dots)$ ¹³. First, we show that X_{-n} is a backward martingale with respect to the defined filtration:

$$\begin{aligned} E[X_{-n}|F_{-n-1}] &= E\left[\frac{S_n}{n} \middle| F_{-n-1}\right] \\ &= \frac{1}{n} \sum_{i=1}^n E[\xi_i | S_{n+1}] \\ &= E[\xi_1 | S_{n+1}] \\ &= \frac{S_{n+1}}{n+1} = X_{-n-1} \end{aligned}$$

By the backward martingale convergence theorem, $X_{-n} \rightarrow X_{-\infty}$ a.s. and in L^1 , where $X_{-\infty} = E[\xi_1 | F_{-\infty}] = E[\xi_1]$.¹⁴ Thus, $\lim_{n \rightarrow \infty} \frac{S_n}{n} = E[\xi_1]$ a.s. $\square$

5 Conclusion

This paper provided proofs of martingales convergence theorems by examining both the **Forward** and the **Backward** versions. The practical power of these results was explored through diverse applications: the forward theorem provided a proof to the **Gambler's Ruin** problem, while the backward theorem derived the **Strong Law of Large Numbers** (SLLN) and the **Hewitt-Savage 0-1 Law**.

References

- [1] R. Durrett. *Probability: Theory and Examples*, Version 5. Department of Mathematics, 2019.
- [2] D. Gamarnik. Massachusetts Institute of Technology¹⁵. Martingale Convergence Theorem (Lecture 11 - Additional material). *6.265/15.070J*, Fall 2013.
- [3] R. Mansuy¹⁶. The Origins of the Word "Martingale". *Journal Electronique d'Histoire des Probabilités et de la Statistique*, Vol. 5, No. 1, 2009.

¹³This is a backward filtration because

$$\mathcal{F}_{-(n+1)} \subset \mathcal{F}_{-n}$$

¹⁴This is proved using the 0-1 Law above

¹⁵This was the main source for the mathematical background discussed in the paper.

¹⁶This was the main source for the introduction

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